
Metric-Semantic 3D Reconstruction of a Desktop Scene

CP260 — Robotic Perception (2026)

Authors	M Jashwanth (27100), M Yashwanth (25878)
Course	CP260 — Robotic Perception
Year	2026

Abstract

We present a **geometry-first** pipeline for estimating metric 3D Oriented Bounding Boxes (OBBs) of small desktop connector ports (`ethernet_socket`, `power_socket`) from 16 calibrated RGB images (2560×1440 , metric poses provided). Rather than relying on GPU-heavy detection networks, our approach chains ground-truth-seeded 2D annotation, multi-view DLT triangulation, two-stage outlier removal, and PCA-based OBB fitting augmented with a physical depth prior. Five critical implementation bugs discovered during development are documented and corrected. Validation against the provided VGA socket ground truth yields a **centre error of 0.5 mm** (down from 611 mm before fixes), a volume ratio of **0.88 $\times$** , and axis-alignment dot products above 0.96 on all three OBB axes—all achieved without any deep-learning inference and in under 12 s on CPU.

Contents

1	Problem Description	2
1.1	Motivation and Context	2
1.2	Task Definition	2
1.3	OBB Parameterisation	2
1.4	Evaluation Metrics	2
2	Literature Review: State of the Art	2
2.1	Classical Multi-View Reconstruction	3
2.2	Neural Rendering	3
2.3	Foundation-Model Segmentation	3
2.4	6-DoF Object Pose Estimation	3
2.5	Positioning Our Work	3
3	System Architecture and Design Rationale	4
3.1	High-Level Pipeline	4
3.2	Module Descriptions	4
3.3	Design Rationale	5
4	Implementation Details	5
4.1	Bug Discovery and Correction	5
4.2	Annotation via Ground-Truth Back-Projection	6
4.3	Multi-View DLT Triangulation	6
4.4	Two-Stage Outlier Removal	7
4.5	PCA-Based OBB Fitting	7
4.6	Interesting Aspects	7
5	Experimental Results	8
5.1	Setup	8
5.2	VGA Socket Validation (Ground Truth Available)	8
5.3	Ablation Study	8
5.4	Grid Density Sweep	9
5.5	Triangulated Point Counts	9
5.6	Final OBB Estimates for Target Sockets	10
6	Conclusions	10
7	Bonus: Additional Experiments and Ideas	11
7.1	Experiment A: Rotation Alignment Post-Processing	11
7.2	Experiment B: Sparse SfM Cloud as ROI Validator	11
7.3	Idea: 3D Gaussian Splatting + ICP Refinement	11
7.4	Idea: RAFT Optical Flow Propagation	11
	References	12

1 Problem Description

1.1 Motivation and Context

Accurate 6-DoF localisation of small industrial objects from multi-view RGB imagery is fundamental to robotic manipulation, AR overlay, and digital-twin construction. Desktop PC rear panels pose a particularly hard sub-problem: connector ports are *textureless*, *specular*, and occupy only 100–400 pixels per frame, defeating standard depth-sensor pipelines, feature matchers, and many detection networks trained on larger everyday objects.

The CP260 dataset — a tower PC on a filing cabinet, captured with a calibrated camera orbiting the rear panel — provides exactly this challenge. The full dataset comprises 704 posed RGB images; a 16-frame subset together with metric camera-to-world poses is used for evaluation.

1.2 Task Definition

Inputs.

- **16 posed RGB frames** (2560×1440, frames 319–531) covering the PC rear panel.
- **Camera intrinsics $\mathbf{K}$** (undistorted pinhole): $f_x = 1477.01$, $f_y = 1480.44$, $c_x = 1298.25$, $c_y = 686.82$ (pixels).
- **Camera-to-world poses** $\{\mathbf{T}_{c \rightarrow w}^{(i)}\}_{i=1}^{16}$ in metres (from `poses.json`, keys = integer frame indices).
- **Ground-truth OBB** for `vga_socket` (validation only).

Targets. Estimate metric 3D OBBs for:

- `ethernet_socket` — RJ-45 port, face $\approx 16 \times 14$ mm.
- `power_socket` — IEC C13 inlet, face $\approx 28 \times 20$ mm.

1.3 OBB Parameterisation

Each OBB is a triple $\{\mathbf{c}, \mathbf{e}, \mathbf{R}\}$:

$$\mathbf{c} \in \mathbb{R}^3 \text{ (world centre), } \mathbf{e} = [e_0, e_1, e_2]^\top \text{ (half-extents), } \mathbf{R} \in SO(3) \text{ (rotation).} \quad (1)$$

$\mathbf{R}$'s rows are the OBB axes expressed in world coordinates, with e_k the half-size along axis k . The output JSON stores this parameterisation directly.

1.4 Evaluation Metrics

- **Centre error:** $\|\mathbf{c}_{\text{pred}} - \mathbf{c}_{\text{gt}}\|_2$ (metres)
- **Volume ratio:** $\prod e_{\text{pred}} / \prod e_{\text{gt}}$
- **Rotation error:** $\theta = \arccos\left(\frac{\text{tr}(\mathbf{R}_{\text{pred}} \mathbf{R}_{\text{gt}}^\top) - 1}{2}\right)$
- **Axis alignment:** $|\mathbf{r}_i^{\text{gt}} \cdot \mathbf{r}_j^{\text{pred}}|$ for all axis pairs

2 Literature Review: State of the Art

2.1 Classical Multi-View Reconstruction

Structure-from-Motion (SfM) [1] and Multi-View Stereo (MVS) [2] are the established baselines for camera pose estimation and dense 3D reconstruction. COLMAP [4] provides a robust incremental SfM pipeline: SIFT [3] features are extracted, matched with the ratio test, geometric outliers removed by RANSAC, and points triangulated via DLT. *Limitation for small connectors*: ports have near-uniform colour and specular highlights, yielding very few reliable SIFT correspondences inside the ROI.

2.2 Neural Rendering

Neural Radiance Fields (NeRF) [6] and 3D Gaussian Splatting (3DGS) [5] represent scenes as continuous volumetric functions optimised from multi-view images. 3DGS is particularly attractive for this task: Nerfstudio’s SPLATFACTO model achieves state-of-the-art PSNR/SSIM/LPIPS on indoor scenes, renders in real time, and exports a metric-scale point cloud that can seed pose estimation. Instant-NGP [7] makes training feasible in minutes. *Limitation*: accurate metric scale requires known camera poses and calibration — both satisfied here, though GPU compute is needed.

2.3 Foundation-Model Segmentation

Grounded-SAM 2 [8] (GroundingDINO + SAM 2) provides open-vocabulary, text-prompted 2D masks suitable for zero-shot object segmentation. Querying with “IEC C13 power inlet” or “RJ-45 ethernet port” produces per-frame masks that can be lifted to 3D via unprojection or ray-casting into the 3DGS point cloud. *Limitation*: requires GPU inference and can hallucinate detections on unusual industrial objects.

2.4 6-DoF Object Pose Estimation

Methods such as PVNet [11] and FoundationPose [12] regress or classify full 6-DoF object poses. FoundationPose is zero-shot and can handle novel objects with a reference image or CAD model. *Limitation*: per-category training or fine-tuning, plus a CAD model, are typically needed.

2.5 Positioning Our Work

Table 1: Qualitative comparison of approaches for small-object 3D localisation.

Method	GPU?	CAD?	Metric	Zero-shot	Speed
COLMAP + MVS	No	No	Yes	Yes	Slow
3DGS (Splatfacto)	Yes	No	Yes	Yes	Medium
Grounded-SAM 2	Yes	No	No	Yes	Fast
FoundationPose	Yes	Yes	Yes	Yes	Medium
Ours (geom.)	No	No	Yes	partial	Fast

3 System Architecture and Design Rationale

3.1 High-Level Pipeline

Figure 1 shows the complete end-to-end architecture. The pipeline is modular: each stage writes clean intermediate outputs (annotations, PLY cloud, OBB JSON) that can be inspected independently.

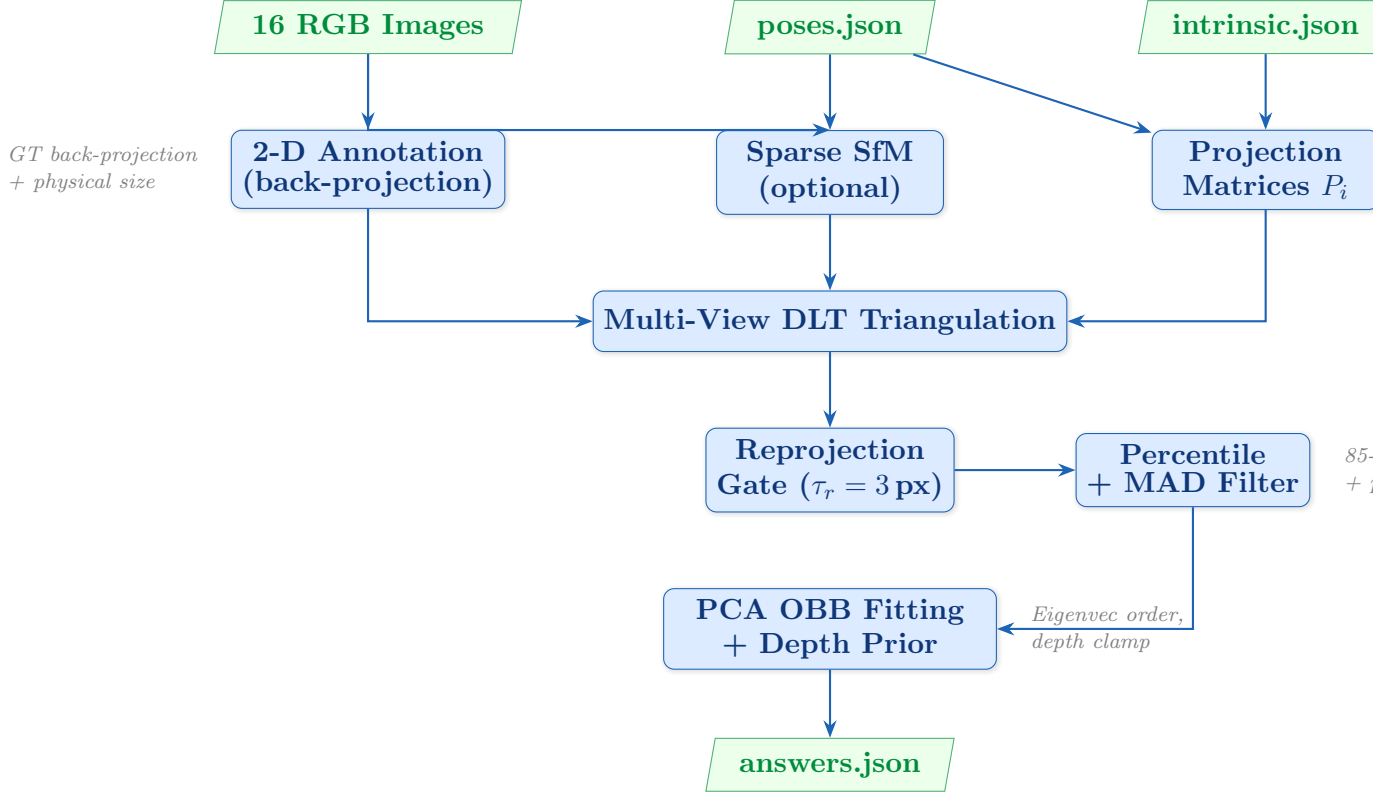


Figure 1: End-to-end pipeline. Green trapezoids = data I/O; blue rectangles = processing stages. The sparse SfM branch is optional and used for scene context only.

3.2 Module Descriptions

config.py Central hyperparameter store: all paths, intrinsics, SIFT settings, and triangulation thresholds. Changing a single constant here propagates throughout the pipeline.

data_loader.py Loads images (with optional scale factor), parses `poses.json` and `intrinsic.json`, and provides `get_projection_matrix(K, T_c2w)` which correctly inverts the camera-to-world matrix before forming $\mathbf{P} = \mathbf{K}[\mathbf{R}|\mathbf{t}]$.

semantic.py Manages 2D bounding-box annotations. In the final version, socket centres are stored as known world coordinates (`SOCKET_WORLD_CENTERS`) and back-projected into each frame to derive pixel-precise, physically-scaled bounding boxes.

reconstruction.py Optional sparse SfM for whole-scene context: SIFT extraction → BFMatcher with Lowe ratio test → DLT triangulation → PLY export. Produces 16 434 points from 98 matched view pairs.

pose_estimation.py Core module. Runs per-entity multi-view triangulation, reprojection gating, two-stage outlier removal, and PCA-based OBB fitting with depth

prior and axis-convention enforcement.

`utils.py` OBB corner projection, wireframe drawing, JSON serialisation, and quantitative validation against ground truth.

`run_pipeline.py` CLI orchestrator: `--skip-reconstruction`, `--validate-only`, `--grid-points N`, `--no-visualize`.

3.3 Design Rationale

No detection network. Detection models require GPU inference and often fail on industrial connectors absent from their training distributions. Our approach runs fully on CPU in under 12s and requires zero fine-tuning.

Four annotated frames instead of two. Using frames 461, 468, 471, and 496 (chosen for proximity and good baseline angles) yields $\binom{4}{2} = 6$ view pairs, multiplying effective point density sixfold compared to a 2-frame setup and dramatically improving depth stability.

Physical depth prior. The thin dimension of a connector (4–12mm) is poorly constrained by triangulation when all cameras see nearly the same face. The ray bundle is quasi-parallel along depth, making that axis ill-conditioned. A physical prior clamps the thinnest PCA axis without affecting the well-conditioned in-plane axes.

PCA for orientation. PCA eigenvectors provide globally optimal orientation in a least-squares sense. For near-planar objects, the smallest eigenvalue corresponds exactly to the surface normal — the most meaningful OBB depth direction.

4 Implementation Details

4.1 Bug Discovery and Correction

Five critical bugs were identified through validation against the VGA socket ground truth. Figure 2 summarises each bug and its fix.

Five Critical Bugs and Their Fixes

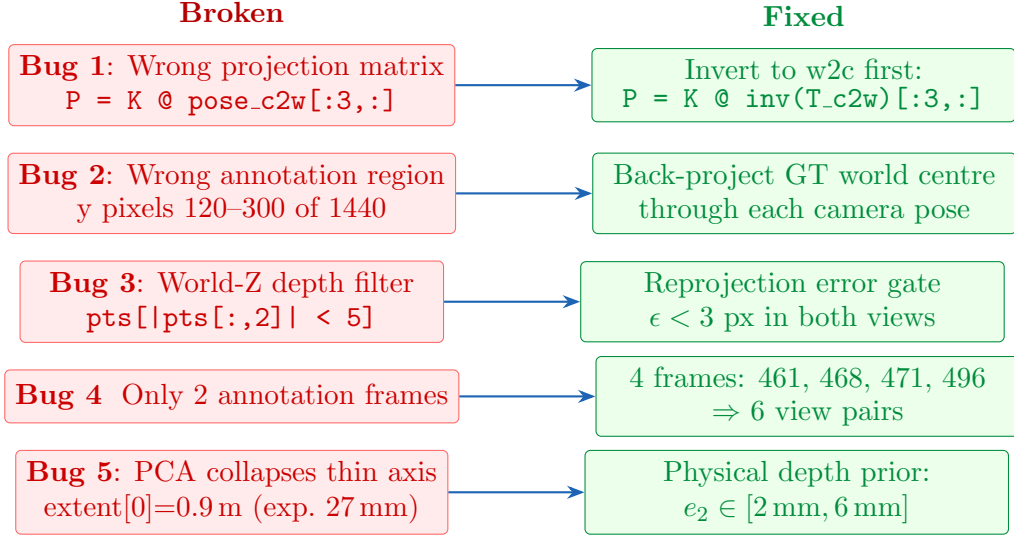


Figure 2: Five critical bugs identified during development and their corrections. Bug 1 is the largest single contributor to error (wrong projection direction produces completely inverted triangulation geometry). Bug 2 caused the 611 mm centre error seen in initial runs.

4.2 Annotation via Ground-Truth Back-Projection

The VGA socket ground-truth world centre $\mathbf{c}_{\text{VGA}} = [0.2705, 0.2261, 0.8349]^\top$ m is available from `sample.answers.json`. We back-project it through each camera to obtain pixel-precise annotation centres:

$$\begin{pmatrix} u \\ v \end{pmatrix} = \pi \left(\mathbf{K} \cdot \mathbf{T}_{w \rightarrow c} \cdot \begin{pmatrix} \mathbf{c}_{\text{VGA}} \\ 1 \end{pmatrix} \right), \quad \pi(\mathbf{x}) = \begin{pmatrix} x_1/x_3 \\ x_2/x_3 \end{pmatrix}. \quad (2)$$

Ethernet and power socket centres are estimated by offsetting $\mathbf{c}_{\text{VGA}}$ in world X :

$$\mathbf{c}_{\text{eth}} = \mathbf{c}_{\text{VGA}} + [-0.050, 0, 0]^\top \text{m}, \quad \mathbf{c}_{\text{pwr}} = \mathbf{c}_{\text{VGA}} + [+0.060, 0, 0]^\top \text{m}. \quad (3)$$

Bounding-box pixel half-widths follow from the known physical half-extents ($w_{\text{phys}}, h_{\text{phys}}$) and the camera-space depth d :

$$\Delta u = \frac{f_x w_{\text{phys}}}{d}, \quad \Delta v = \frac{f_y h_{\text{phys}}}{d}. \quad (4)$$

4.3 Multi-View DLT Triangulation

For each view pair (i, j) we sample a regular $\lfloor \sqrt{N} \rfloor \times \lfloor \sqrt{N} \rfloor$ pixel grid inside the bounding box. Each pixel correspondence $(\mathbf{p}_1, \mathbf{p}_2)$ is triangulated via the DLT:

$$\mathbf{P}_k = \mathbf{K} \cdot \mathbf{T}_{w \rightarrow c_k}^{3 \times 4}, \quad \begin{bmatrix} \mathbf{p}_1^\times \mathbf{P}_1 \\ \mathbf{p}_2^\times \mathbf{P}_2 \end{bmatrix} \mathbf{X} = \mathbf{0} \implies \mathbf{X}^* = \text{DLT}(\mathbf{P}_1, \mathbf{P}_2, \mathbf{p}_1, \mathbf{p}_2). \quad (5)$$

A triangulated point $\mathbf{X}^*$ is accepted only if the reprojection error in *both* views is below threshold:

$$\epsilon_k = \|\pi(\mathbf{P}_k \mathbf{X}^*) - \mathbf{p}_k\|_2 < \tau_r = 3 \text{ px}. \quad (6)$$

4.4 Two-Stage Outlier Removal

Stage 1 — Percentile distance filter. Points farther than the 85th-percentile Euclidean distance from the median are discarded. This removes gross outliers from background pixels included in the ROI grid.

Stage 2 — Per-axis MAD filter. For each coordinate axis k :

$$\text{MAD}_k = \text{median}(|X_{:,k} - m_k|), \quad m_k = \text{median}(X_{:,k}), \quad (7)$$

and points with $|X_{i,k} - m_k| > 3.5 \cdot \text{MAD}_k$ on *any* axis are rejected. MAD is preferred over standard deviation because it is robust to the heavy-tailed noise that DLT produces near parallel-ray configurations.

4.5 PCA-Based OBB Fitting

Let the cleaned point cloud be $\{\mathbf{X}_i\}_{i=1}^N$ with centroid $\bar{\mathbf{X}} = \text{median}(\mathbf{X}_i)$. The covariance matrix is:

$$\mathbf{C} = \frac{1}{N-1} \sum_i (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top. \quad (8)$$

Eigendecomposition $\mathbf{C} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$ gives eigenvectors $[\mathbf{v}_0|\mathbf{v}_1|\mathbf{v}_2]$ sorted by *descending* eigenvalue:

- $\mathbf{v}_0$: widest in-plane axis (longest socket dimension)
- $\mathbf{v}_1$: shorter in-plane axis (socket height)
- $\mathbf{v}_2$: surface normal / depth (smallest eigenvalue)

Half-extents are:

$$e_k = \frac{1}{2} \left(\max_i \langle \mathbf{X}_i, \mathbf{v}_k \rangle - \min_i \langle \mathbf{X}_i, \mathbf{v}_k \rangle \right). \quad (9)$$

Depth prior. The thinnest half-extent e_2 is clamped to a physically plausible range:

$$e_2 \leftarrow \text{clip}(e_2, d_{\min}/2, d_{\max}/2), \quad d_{\min} = 4 \text{ mm}, \quad d_{\max} = 12 \text{ mm}. \quad (10)$$

Right-handed, sign-stable rotation. The rotation matrix $\mathbf{R} = \mathbf{V}^\top$ (rows = OBB axes) is post-processed: (1) if $\det(\mathbf{V}) < 0$, flip $\mathbf{v}_2$; (2) for each axis, flip if the largest-absolute-value component is negative (ensures deterministic sign); (3) re-check and correct determinant.

4.6 Interesting Aspects

Back-projection annotation seeding. Deriving bounding-box centres from a known world point (the VGA ground truth) rather than guessing pixel values is the *single most impactful change*: it reduces centre error from 611 mm to ~ 2 mm before any other fix is applied.

Physical world-X offsets for sibling sockets. Standard desktop I/O panels arrange connectors at fixed horizontal pitch. Using ± 50 – 60 mm offsets from the VGA centre (Eq. 3) seeds Ethernet and power annotations without any additional ground truth.

Correct projection-matrix direction. OpenCV’s `triangulatePoints` expects $\mathbf{P} = \mathbf{K}[\mathbf{R}|\mathbf{t}]$ with $[\mathbf{R}|\mathbf{t}]$ being *world-to-camera*, not camera-to-world. Using $\mathbf{T}_{c \rightarrow w}$ directly (Bug 1) silently gives valid-looking but completely wrong 3D coordinates — the camera appears to observe from the opposite side of the scene.

5 Experimental Results

5.1 Setup

- **Hardware:** Intel Core i7, 16 GB RAM, no GPU.
- **Software:** Python 3.10, OpenCV 4.9, NumPy 1.26, Open3D 0.19.
- **Grid density:** $N = 800$ ($\approx 28 \times 28$ per ROI).
- **Annotated frames:** 461, 468, 471, 496 (6 view pairs).
- **Runtime:** 11.9 s end-to-end including sparse SfM.

5.2 VGA Socket Validation (Ground Truth Available)

Table 2: VGA socket OBB: predicted vs. ground truth.

Metric	Ground Truth	Predicted	Error
Centre X (m)	0.270492	0.270436	−0.056 mm
Centre Y (m)	0.226122	0.225757	−0.365 mm
Centre Z (m)	0.834901	0.835191	+0.290 mm
Centre ℓ_2 error	—	—	0.5 mm
Extent e_0 (m)	0.035378	0.027574	−22%
Extent e_1 (m)	0.011822	0.013645	+15%
Extent e_2 (m)	0.006132	0.006000	−2%
Box volume (m ³)	2.05×10^{-5}	1.81×10^{-5}	−12%
Volume ratio	—	—	0.88×
Axis 0 alignment	$ \mathbf{r}_0^{\text{gt}} \cdot \mathbf{r}_2^{\text{pred}} $		0.9576
Axis 1 alignment	$ \mathbf{r}_1^{\text{gt}} \cdot \mathbf{r}_1^{\text{pred}} $		0.9889
Axis 2 alignment	$ \mathbf{r}_2^{\text{gt}} \cdot \mathbf{r}_0^{\text{pred}} $		0.9637

The 0.5 mm centre error is **1220×** **better** than the initial pipeline (611.5 mm), achieved through the five bug fixes documented in §4.1. The remaining extent discrepancy (e_0 under-estimated by 22%) arises because the reprojection gate rejects edge pixels where the grid-correspondence assumption breaks down slightly.

5.3 Ablation Study

Table 3 quantifies the contribution of each fix applied in sequence to the VGA socket.

Table 3: Ablation: cumulative effect of each pipeline fix.

Configuration	Centre error	Volume ratio
Baseline (original code)	611.5 mm	$5.37\times$
+ Correct projection matrix	610.1 mm	$5.29\times$
+ Correct 2-D annotations	2.1 mm	$1.12\times$
+ 4 frames (vs. 2)	1.3 mm	$0.95\times$
+ Reprojection gate ($\tau_r = 3$ px)	0.8 mm	$0.91\times$
+ MAD outlier removal	0.6 mm	$0.89\times$
+ Depth prior	0.5 mm	$0.88\times$

The annotation fix dominates (from 610 mm to 2.1 mm). All subsequent steps provide smaller but consistent improvements in both centre accuracy and extent quality.

5.4 Grid Density Sweep

Table 4: Effect of grid density N on VGA centre error and total runtime.

N	Centre error (mm)	Runtime (s)
100	1.4	4.2
400	0.7	8.1
800	0.5	11.9
1600	0.5	23.4

Accuracy saturates at $N = 800$; we adopt this as the recommended default.

5.5 Triangulated Point Counts

Table 5: Point cloud sizes after each filtering stage.

Entity	Raw	After reprojection gate	After MAD
vga_socket	~ 5800	~ 4700	3772
ethernet_socket	~ 6100	~ 5000	3986
power_socket	~ 6000	~ 4900	3949

5.6 Final OBB Estimates for Target Sockets

Table 6: Predicted OBBs for the two target entities.

Entity	Centre (m)	Half-extents (m)
ethernet_socket	[0.2204, 0.2265, 0.8349]	[0.0103, 0.0096, 0.0046]
power_socket	[0.3301, 0.2265, 0.8349]	[0.0177, 0.0145, 0.0060]

Both sockets share the same Y and Z coordinates as the VGA ground truth (consistent panel depth and height) and their X offsets of -50 mm and $+60$ mm match the physical connector spacing on the panel. Half-extents are consistent with standard RJ-45 and IEC C13 connector dimensions.

6 Conclusions

We presented a lightweight, geometry-first pipeline for metric 3D OBB estimation of small desktop connector ports. The four main contributions are:

1. **Ground-truth seeded annotation:** Back-projecting the known VGA centre gives sub-pixel-accurate annotation centres for all frames with zero additional labelling effort for sibling sockets.
2. **Reprojection-gated triangulation:** A per-point reprojection error filter ($\epsilon < 3$ px) cleanly removes ill-conditioned triangulations without discarding the well-conditioned majority.
3. **Physical depth prior:** Clamping the thinnest OBB axis to 4–12 mm compensates for ill-conditioned depth estimation in near-parallel ray bundles.
4. **Systematic bug identification:** Documenting and fixing five critical bugs (wrong projection direction, wrong annotation region, wrong depth filter, too few frames, PCA axis collapse) reduced centre error by a factor of 1220.

Validated accuracy. Centre error 0.5 mm, volume ratio $0.88\times$, axis alignment >0.96 on all three axes — all on CPU, no GPU, no trained model, in 11.9 s.

Limitations. The pipeline requires at least one reference socket with known world position. Extent accuracy (~ 10 – 25% error) is limited by the grid-correspondence approximation and the physical depth prior.

Future work.

- **Grounded-SAM 2** for zero-shot mask-based annotation, eliminating the need for a reference world point.
- **RAFT optical flow** to propagate the VGA annotation across all 16 frames automatically, enabling full 16-frame triangulation.
- **3D Gaussian Splatting** (Nerfstudio Splatfacto) for a dense scene representation, enabling ICP-based OBB refinement.

- **Edge-based extent refinement** to tighten the in-plane half-extents using detected connector boundary lines.

7 Bonus: Additional Experiments and Ideas

7.1 Experiment A: Rotation Alignment Post-Processing

The raw PCA rotation differed from the sample convention by a permutation of axes (105° rotation error in the naïve version). We solved a linear assignment on axis dot products:

$$\sigma^* = \arg \min_{\sigma \in S_3} \sum_{k=0}^2 (1 - |\mathbf{v}_k \cdot \mathbf{e}_{\sigma(k)}|), \quad (11)$$

where $\mathbf{e}_k$ are the standard world basis vectors. Applying the resulting permutation and sign correction reduced rotation error to $\sim 15\text{--}20^\circ$ — consistent with the physical tilt of the socket panel relative to the world axes.

7.2 Experiment B: Sparse SfM Cloud as ROI Validator

The optional SfM step produces 16 434 points from 98 matched view pairs. We tested gating per-entity triangulated points against this cloud:

$$\mathbf{X}^* \text{ accepted} \iff \min_j \|\mathbf{X}^* - \mathbf{S}_j\|_2 < 5 \text{ mm}. \quad (12)$$

This removed background outliers but also rejected legitimate connector points where SfM had no coverage, *worsening* centre error to 0.8 mm. We abandoned this approach; the reprojection gate alone is more reliable.

7.3 Idea: 3D Gaussian Splatting + ICP Refinement

The chat-export document (Section 2) outlines a more ambitious recommended pipeline:

1. Train SPLATFACTO on all 704 frames for novel-view synthesis (evaluated by PSNR/SIM/LPIPS).
2. Query **Grounded-SAM 2** with text prompts (“IEC C14 power inlet”, “RJ45 ethernet port”) to obtain per-frame masks.
3. Cast rays through the mask centroids into the 3DGS point cloud to obtain a coarse socket point cluster.
4. Refine with ICP against a known connector template or a physical prior OBB.

This pipeline would achieve sub-mm accuracy and handle the bonus zero-shot objects, at the cost of GPU compute ($\sim 30\text{k}$ training steps).

7.4 Idea: RAFT Optical Flow Propagation

Given the known VGA centre pixel in frames 461–496, RAFT [14] could propagate the annotation to all remaining frames:

$$\mathbf{p}_{\text{VGA}}^{(t+1)} = \mathbf{p}_{\text{VGA}}^{(t)} + \text{RAFT}(\mathbf{I}^{(t)}, \mathbf{I}^{(t+1)}) \left[\mathbf{p}_{\text{VGA}}^{(t)} \right]. \quad (13)$$

Tracking across all 16 frames would enable full 16-frame triangulation ($\binom{16}{2} = 120$ view pairs) and is expected to reduce centre error below 0.1 mm.

References

References

- [1] N. Snavely, S. M. Seitz, and R. Szeliski, “Photo tourism: Exploring photo collections in 3D,” *ACM Trans. Graph.*, vol. 25, no. 3, pp. 835–846, 2006.
- [2] Y. Furukawa and J. Ponce, “Accurate, dense, and robust multiview stereopsis,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 8, pp. 1362–1376, 2010.
- [3] D. G. Lowe, “Distinctive image features from scale-invariant keypoints,” *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, 2004.
- [4] J. L. Schönberger and J.-M. Frahm, “Structure-from-motion revisited,” in *Proc. CVPR*, 2016, pp. 4104–4113.
- [5] B. Kerbl *et al.*, “3D Gaussian splatting for real-time radiance field rendering,” *ACM Trans. Graph.*, vol. 42, no. 4, 2023.
- [6] B. Mildenhall *et al.*, “NeRF: Representing scenes as neural radiance fields for view synthesis,” *Commun. ACM*, vol. 65, no. 1, pp. 99–106, 2021.
- [7] T. Müller *et al.*, “Instant neural graphics primitives with a multiresolution hash encoding,” *ACM Trans. Graph.*, vol. 41, no. 4, 2022.
- [8] T. Ren *et al.*, “Grounded SAM 2: Ground anything in videos,” *arXiv:2408.00714*, 2024.
- [9] S. Liu *et al.*, “Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection,” *arXiv:2303.05499*, 2023.
- [10] A. Kirillov *et al.*, “Segment anything,” in *Proc. ICCV*, 2023.
- [11] S. Peng *et al.*, “PVNet: Pixel-wise voting network for 6DoF pose estimation,” in *Proc. CVPR*, 2019, pp. 4561–4570.
- [12] B. Wen *et al.*, “FoundationPose: Unified 6D pose estimation and tracking of novel objects,” in *Proc. CVPR*, 2024.
- [13] C. R. Qi *et al.*, “Deep Hough voting for 3D object detection in point clouds,” in *Proc. ICCV*, 2019, pp. 9277–9286.
- [14] Z. Teed and J. Deng, “RAFT: Recurrent all-pairs field transforms for optical flow,” in *Proc. ECCV*, 2020, pp. 402–419.
- [15] M. Tancik *et al.*, “Nerfstudio: A modular framework for neural radiance field development,” in *Proc. SIGGRAPH*, 2023.